

DATA

PREPROCESSING – PART II

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PREPROCESSING

The following concepts will be introduced:

- ✓ **CURSE OF DIMENSIONALITY**
- ✓ **DIMENSIONALITY REDUCTION**
- ✓ **BINARIZATION/DISCRETIZATION**
- ✓ **VARIABLE TRANSFORMATION**

In many cases we have to analyze data sets characterized by an high number of attributes.

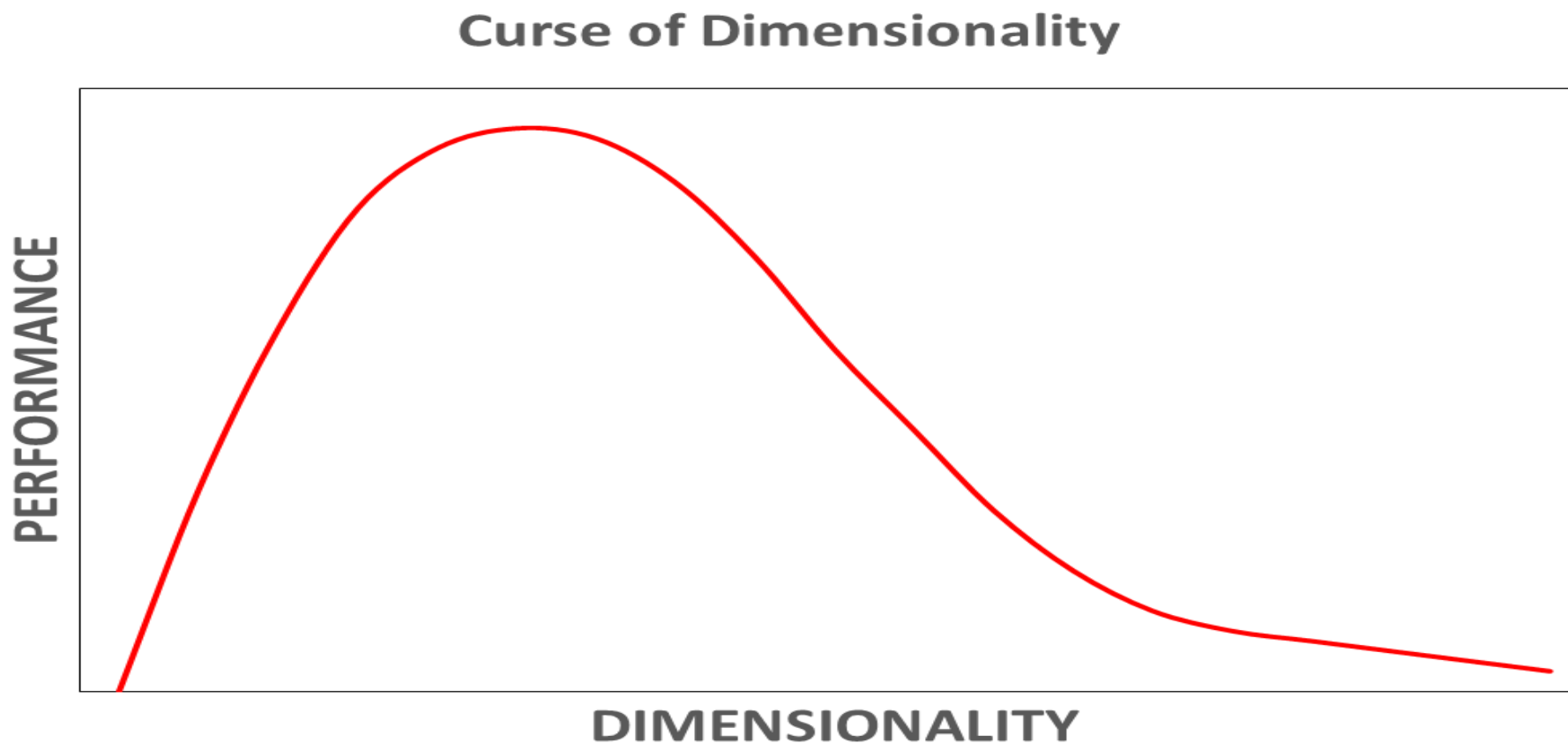
DOCUMENTS REPRESENTED AS WORDS' FREQUENCY, WORDS ARE FROM A VOCABULARY WHICH EASILY CONTAINS TEN OF THOUSANDS OF ELEMENTS (ATTRIBUTES).

REDUCING THE NUMBER OF ATTRIBUTES HAS SEVERAL ADVANTAGES

- ✓ many **DATA MINING ALGORITHMS WORK BETTER** if the dimensionality (number of attributes) is lower (irrelevant attributes are removed while noise in data is reduced).
- ✓ **INTERPRETABILITY** of the developed model is **INCREASED**, it depends on a lower number of attributes.
- ✓ **GRAPHICAL REPRESENTATION** of data is facilitated.
- ✓ **AMOUNT OF TIME AND MEMORY** required by data mining algorithms is reduced.

Many types of **DATA ANALYSIS** become significantly **HARDER AS** the **DIMENSIONALITY OF THE DATA INCREASES**.

As **DIMENSIONALITY INCREASES**, the **DATA BECOMES INCREASINGLY SPARSE** in the space it occupies.



Some of the most common approaches for dimensionality reduction, particularly for **CONTINUOUS ATTRIBUTES**, use **TECHNIQUES FROM LINEAR ALGEBRA TO PROJECT THE DATA FROM A HIGH-DIMENSIONAL SPACE INTO A LOWER-DIMENSIONAL SPACE**.

PRINCIPAL COMPONENT ANALYSIS (PCA), **FINDS** new attributes (**PRINCIPAL COMPONENTS**) that:

1. are **LINEAR COMBINATIONS OF THE ORIGINAL ATTRIBUTES**
2. are **ORTHOGONAL** (perpendicular) **TO EACH OTHER**
3. **CAPTURE THE MAXIMUM AMOUNT OF VARIATION** in the data

We are usually asked to **SPECIFY THE NUMBER OF PRINCIPAL COMPONENTS TO RETAIN** or the **PERCENTAGE OF VARIATION** we want **TO EXPLAIN**.

SINGULAR VALUE DECOMPOSITION (SVD) is a linear algebra technique that is related to PCA and it is also commonly used for dimensionality reduction.

It may be useful to **TRANSFORM CONTINUOUS AND DISCRETE ATTRIBUTES INTO ONE OR MORE BINARY ATTRIBUTES**.

Such a procedure is called **BINARIZATION**.

Assume you have a data set where the value of the **QUALITATIVE ATTRIBUTE** named **TASTE** measures the **CUSTOMER'S JUDGEMENT OF A NEW TYPE OF CANNED SOUP**.



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Taste	Integer Value
awful	0
poor	1
ok	2
good	3
great	4

Associate to the **5** ($k=5$) possible **VALUES** that **TASTE CAN TAKE** on, **5 INTEGER VALUES IN THE INTERVAL $[0,4]$** ($[0,k-1]$).

If the **ATTRIBUTE IS ORDINAL**, then **THE ORDER MUST BE MAINTAINED BY THE ASSIGNMENT**.

This transformation is **REQUIRED ALSO IF THE ATTRIBUTE IS REPRESENTED BY INTEGERS**, in the case where such **INTEGERS ARE NOT IN THE INTERVAL $[0,k-1]$** .

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CONVERT the **5** (k) **INTEGERS TO A BINARY NUMBER.**

$$s = \lceil \log_2 k \rceil$$

BINARY DIGITS TO REPRESENT K INTEGERS.

S=3, THUS 3 BINARY ATTRIBUTES ARE REQUIRED TO REPRESENT AN ATTRIBUTE WHICH CAN TAKE 5 INTEGER VALUES.

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Taste	Integer Value	X_1	X_2	X_3
awful	0	0	0	0
poor	1	0	0	1
ok	2	0	1	0
good	3	0	1	1
great	4	1	0	0

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It may be the case that **ONLY THE PRESENCE OF THE VALUE 1 FOR A BINARY ATTRIBUTE IS IMPORTANT**.

MARKET BASKET ANALYSIS, ONLY ITEMS THAT ARE INCLUDED IN THE CUSTOMER'S BASKET ARE IMPORTANT.

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Taste	Integer Value	X_1	X_2	X_3	X_4	X_5
awful	0	1	0	0	0	0
poor	1	0	1	0	0	0
ok	2	0	0	1	0	0
good	3	0	0	0	1	0
great	4	0	0	0	0	1

It is necessary to **INTRODUCE ONE BINARY ATTRIBUTE FOR EACH VALUE THAT THE CATEGORICAL ATTRIBUTE CAN TAKE ON**.

Typically **APPLIED TO ATTRIBUTES** that are **USED IN CLASSIFICATION OR ASSOCIATION ANALYSIS**.

The **BEST DISCRETIZATION DEPENDS ON THE ALGORITHM BEING USED**, as well as other attributes to be considered.

Typically, the **DISCRETIZATION OF AN ATTRIBUTE IS CONSIDERED IN ISOLATION**.



**WHERE TO LOCATE
THE SPLIT POINTS**

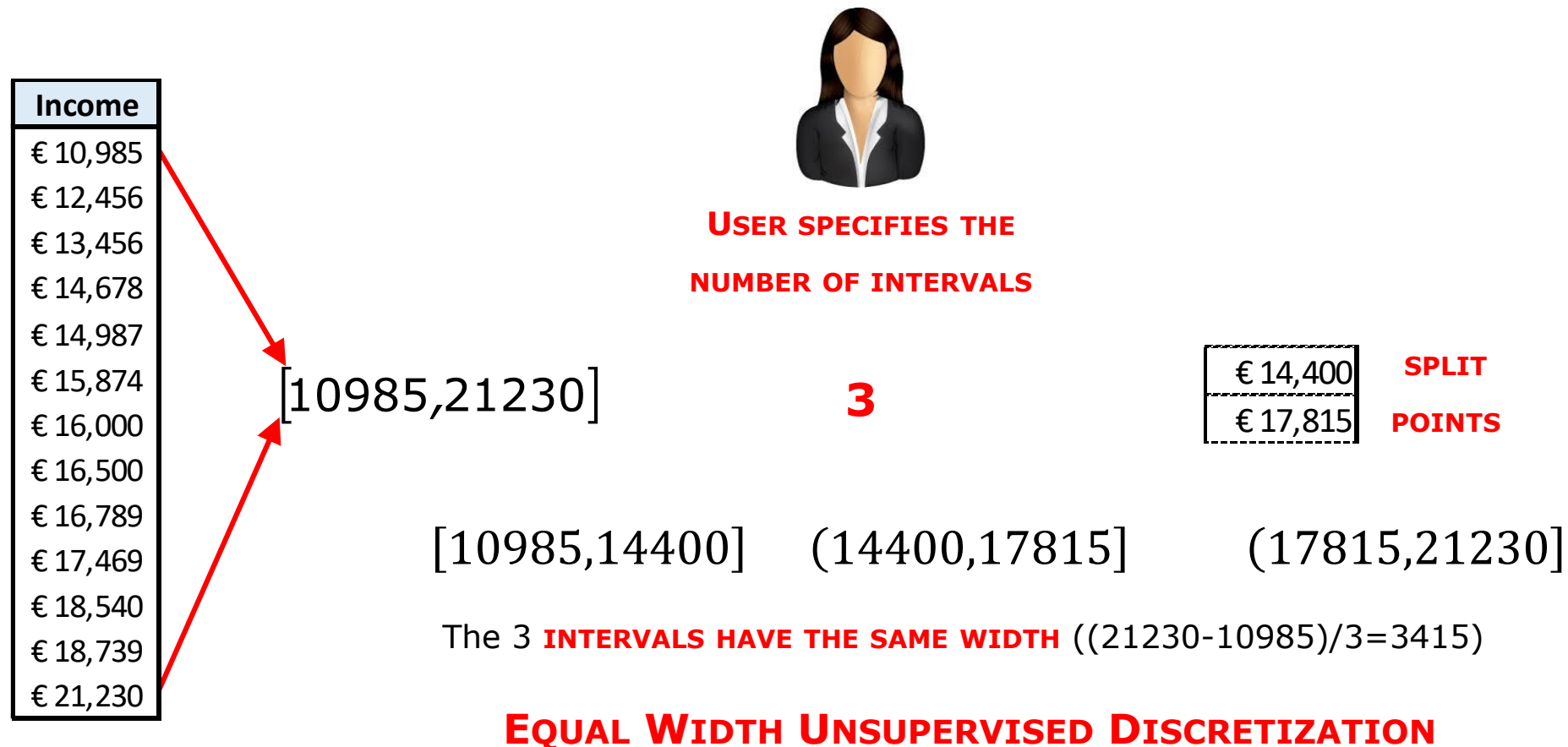


**HOW MANY
CATEGORIES TO HAVE**

Income		Income
€ 10,985	← sort →	€ 15,874
€ 12,456		€ 21,230
€ 13,456		€ 18,739
€ 14,678		€ 16,500
€ 14,987		€ 13,456
€ 15,874		€ 18,540
€ 16,000		€ 17,469
€ 16,500		€ 12,456
€ 16,789		€ 10,985
€ 17,469		€ 14,678
€ 18,540		€ 14,987
€ 18,739		€ 16,000
€ 21,230		€ 16,789

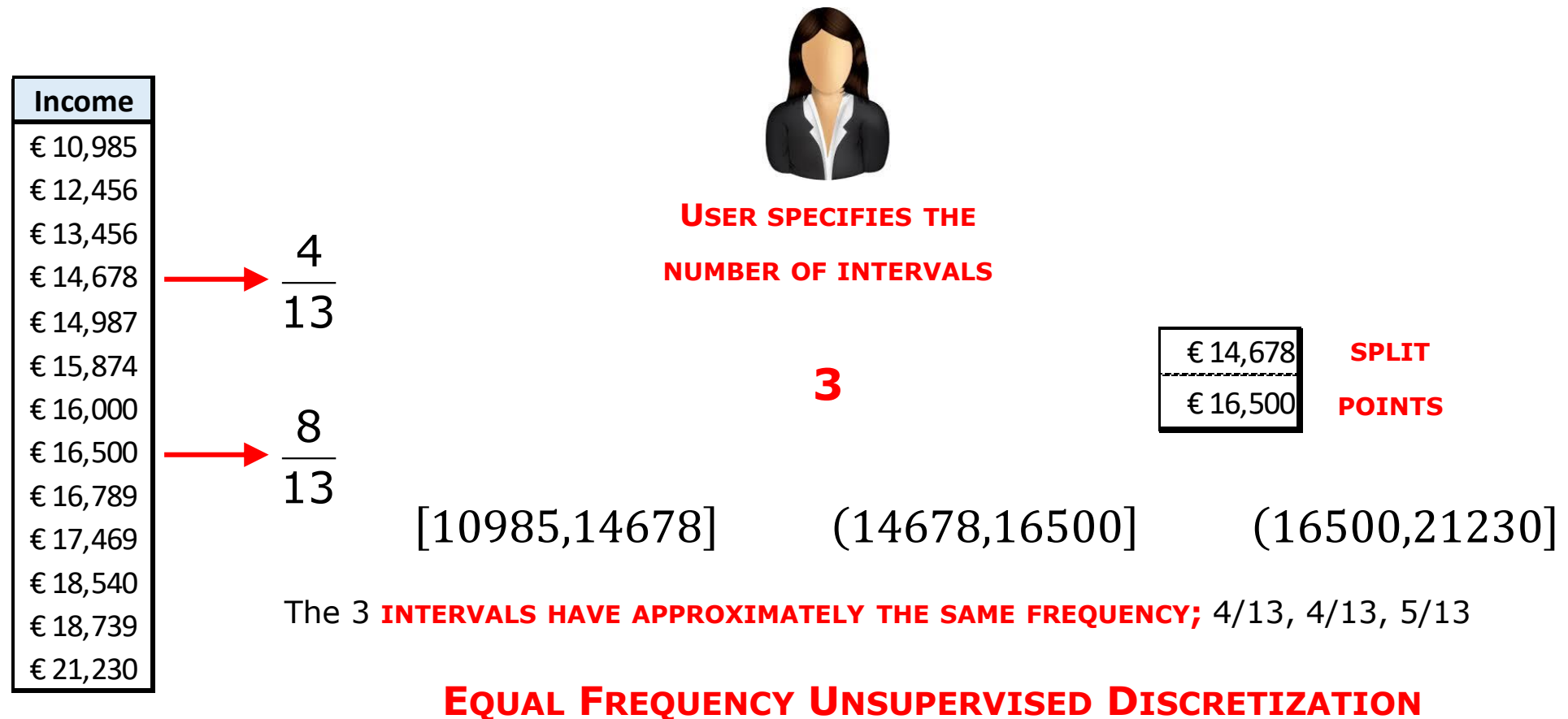
DISCRETIZATION can be **UNSUPERVISED OR SUPERVISED**.

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DISCRETIZATION can be **UNSUPERVISED OR SUPERVISED**.

✓ **SUPERVISED DISCRETIZATION** exploits additional information (**CLASS ATTRIBUTE**) to discretize the Continuous Attribute.

SUPERVISED DISCRETIZATION places split points in such way that some **MEASURE OF PURITY** of the resulting intervals **IS MAXIMIZED**, the **PURITY MEASURE** is computed exploiting **THE CLASS ATTRIBUTE**.

ENTROPY is usually computed as a **MEASURE OF PURITY OF AN INTERVAL**:

$$e_i = -\sum_{k=1}^K p_{ki} \cdot \log_2(p_{ki})$$

ENTROPY associated with the i-th interval, if it

- contains only records of a given class, then $e_i = 0$ maximum purity
- contains equally often all classes, then e_i is maximum minimum purity

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SUPERVISED DISCRETIZATION BASED ON ENTROPY aims to **FIND THE SPLIT POINTS OF THE CONTINUOUS ATTRIBUTE SUCH THAT THE OVERALL ENTROPY IS MINIMIZED** (purity is maximized).

n = number of intervals

m = number of records

m_i = number of records in interval i

$$E = \sum_{i=1}^n w_i \cdot e_i$$

$$w_i = \frac{m_i}{m}$$

CATEGORICAL ATTRIBUTES can sometimes have **TOO MANY VALUES**.

- ✓ If the **CATEGORICAL ATTRIBUTE IS ORDINAL**, then **TECHNIQUES SIMILAR TO THOSE FOR CONTINUOUS ATTRIBUTES** can be used to reduce the number of categories.
- ✓ If the **CATEGORICAL ATTRIBUTE IS NOMINAL**, then **OTHER APPROACHES ARE NEEDED**.

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	?	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	n	7	1.78	?	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	y	?	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA		355-9993
147	0	157	y	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173

Your friend informs you that **STATES ATTRIBUTE VALUES ARE GROUPED INTO STATES CATEGORY**.

Thus, you take the decision to create a **NEW ATTRIBUTE** that you name **STATE_CAT** whose value **DEPENDS ON** the value of the **STATE ATTRIBUTE**.

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128	25	265.1	?	3	2.7	KS	C1
107	26	161.6	n	3	3.7	OH	C2
137	0	243.4	n	5	3.29	NJ	C1
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The **STATE_CAT ATTRIBUTE TAKES VALUE ON A SMALLER SET THAN STATE**:

$$\text{STATE_CAT} \in \{C1, C2, C3, C4\}$$

You **EXPLOITED DOMAIN KNOWLEDGE** and generated the **NEW ATTRIBUTE** named **STATE_CAT**.

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WHAT TO DO WHEN DOMAIN KNOWLEDGE IS NOT AVAILABLE?



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EMPIRICAL APPROACH, such as **GROUPING VALUES** together only **IF** such **GROUPING** results in **IMPROVED** classification **PERFORMANCE** or **ACHIEVES** some other **DATA MINING OBJECTIVE**.

A **VARIABLE TRANSFORMATION** refers to a **TRANSFORMATION** that is **APPLIED TO ALL THE VALUES** of a variable.

Two **TYPES OF VARIABLE TRANSFORMATIONS**:

- ✓ **SIMPLE FUNCTIONS**; a simple mathematical function is applied to each value individually.



logarithm, square root, trigonometric functions, ...

- *Does the order need to be maintained?*
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- ✓ **NORMALIZATION OR STANDARDIZATION**; transforms entire set of values to have a particular property.

X has mean equal to μ and standard deviation equal to σ

$$Z = \frac{X - \mu}{\sigma}$$

Z has mean equal to 0 and standard deviation equal to 1

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SUM OF DIFFERENT CONTINUOUS ATTRIBUTES, avoids one or **FEW ATTRIBUTES TAKING LARGE VALUES** to **DOMINATE** the new attribute **SUM**. The same applies to other possibilities to combine attributes.

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ESTIMATORS of **MEAN and STANDARD DEVIATION** are **STRONGLY AFFECTED BY ANOMALOUS OBSERVATIONS (OUTLIERS)** so the **STANDARDIZATION** is often **MODIFIED**.

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MEAN replaced by **MEDIAN**

STANDARD DEVIATION replaced by **ABSOLUTE STANDARD DEVIATION**
OR
(ABSOLUTE AVERAGE DEVIATION)

X = attribute
 x_i = value of X for the i^{th} record
 μ = mean of X
 m = number of records

$$\sigma_X = \frac{1}{m} \sum_{i=1}^m |x_i - \mu|$$